Name: _____

WORKSHEET

Aerial and nautical applications

1 For each of the following situations:

i draw a sketch of the situation, where **h** is the heading, **c** is the current and **t** is the speed and course made good.

ii write **h** in terms of **i** and **j**.

iii write **c** in terms of **i** and **j**.

iv find **t** using vector addition.

v find $|\mathbf{t}|$ (the speed made good or the ground speed).

vi find the angles in the triangle from your sketch opposite **t** and **c**.

vii find the direction of **t** (the course made good or the true course).

viii state the speed and course made good.

a A yacht sailing 11 knots through the water is headed on a bearing of 159°. There is a current of 3 knots in a direction of 312°.

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b A boat making 4.5 knots through the water is headed on a bearing of 56°. There is a current of 2 knots in a direction of 155°.

c A plane is headed on a bearing of 212° at an airspeed of 100 knots. There is a 10 knot north-westerly wind.

d A plane is headed on a bearing of 345° at an airspeed of 120 knots. There is an 18 knot southerly wind.

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2 For each of the following situations:

i draw a sketch of the situation.

ii find the speed made good/ground speed.

iii find the course made good/true course.

a A yacht is sailing on a bearing of 22° at 9 knots through a south-easterly current of 1.5 knots.

b A plane is headed on a north-westerly bearing at an airspeed of 110 knots through a 15 knot wind northerly wind.

c A cruise ship is sailing on a bearing of 119° at 6 knots through a current of 1.5 knots at 222°.

d A plane is headed on a bearing of 215° at an airspeed of 120 knots through a 10 knot wind blowing at 255°.

3 To solve each of the following problems, you will need to draw a sketch.

a A pilot is flying at 125 knots on a heading of 350°. There is a wind of 11 knots blowing at 55°. Find her true course and groundspeed.

b A captain is sailing at 15 knots on a heading of 300° against a strong current of 4 knots at 140°. Find her speed and course made good.

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c A pilot wants to fly at 125 knots (groundspeed) on a true course of 200°. There is a south-easterly wind of 11 knots. Find the heading and airspeed he should make.

d A captain wants to sail at 20 knots on a true course of 115°. There is a current of 3 knots in a direction of 215°. Find the heading she should make and her speed made good.

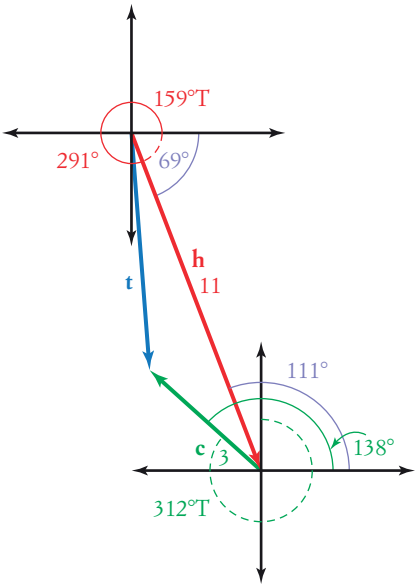
e A captain is sailing at a heading of 75° at a speed made good of about 12 knots. There is a south-westerly current of 2 knots. What is his true course and his speed through water?

f A pilot is flying at 150 knots on a heading of 160°. He is recorded as having a ground speed of 138 knots on a true course of 165°T. What is the speed and direction of the wind?

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Answers

1 a i



ii $\mathbf{h} = 11\cos(291°)\,\mathbf{i} + 11\sin(291°)\,\mathbf{j} \approx 3.94\mathbf{i} - 10.27\mathbf{j}$.

iii $\mathbf{c} = 3\cos(138°)\,\mathbf{i} + 3\sin(138°)\,\mathbf{j} \approx -2.23\mathbf{i} + 2.01\mathbf{j}$.

iv $\mathbf{t} = \mathbf{h} + \mathbf{c} = 1.71\mathbf{i} - 8.26\mathbf{j}$

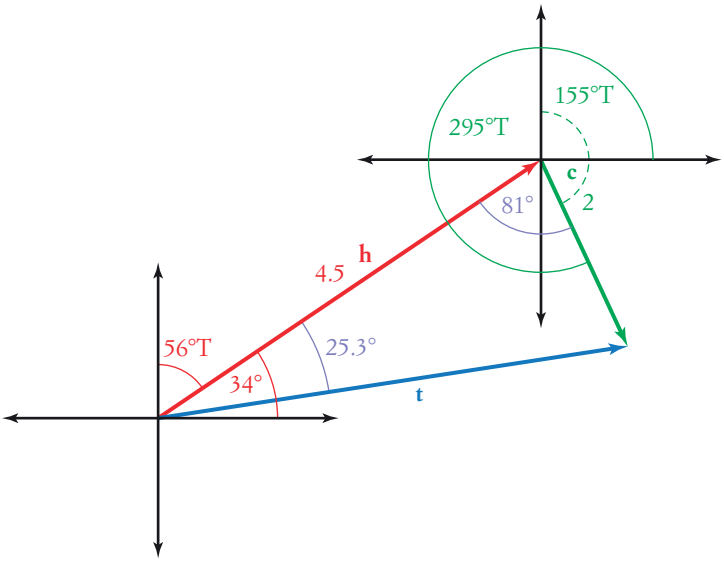
v $|\mathbf{t}| = \sqrt{1.71^2 + (-8.26)^2} \approx 8.44$

vi $\mathbf{T} = 27°$ and $\mathbf{C} = 9.29°$

vii $\mathbf{t}$ is on a bearing of 168.3°

viii The speed and course made good is about 8.44 knots on a bearing of 168.3°.

b i



ii $\mathbf{h} = 4.5\cos(34°)\,\mathbf{i} + 4.5\sin(34°)\,\mathbf{j} \approx 3.73\mathbf{i} + 2.52\mathbf{j}$.

iii $\mathbf{c} = 2\cos(295°)\,\mathbf{i} + 2\sin(295°)\,\mathbf{j} \approx 0.85\mathbf{i} - 1.81\mathbf{j}$.

iv $\mathbf{t} = \mathbf{h} + \mathbf{c} = 4.58\mathbf{i} + 0.70\mathbf{j}$

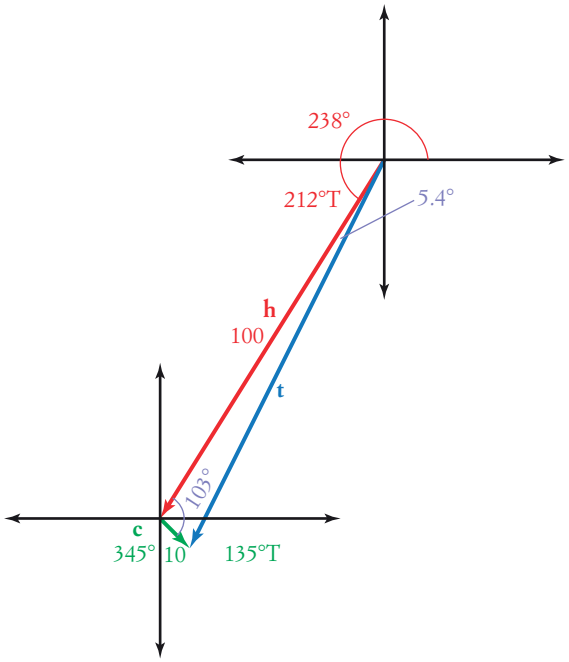
v $|\mathbf{t}| = \sqrt{4.58^2 + 0.70^2} \approx 4.63$

vi $\mathbf{T} = 81°$ and $\mathbf{C} = 25.3°$

vii **t** is on a bearing of 81.3°

viii The speed and course made good is about 4.63 knots on a bearing of 81.3°.

c i



ii $\mathbf{h} = 100\cos(238°)\,\mathbf{i} + 100\sin(238°)\,\mathbf{j} \approx -52.99\mathbf{i} - 84.80\mathbf{j}$.

iii $\mathbf{c} = 10\cos(315°)\,\mathbf{i} + 10\sin(315°)\,\mathbf{j} \approx 7.07\mathbf{i} - 7.07\mathbf{j}$.

iv $\mathbf{t} = \mathbf{h} + \mathbf{c} = -45.92\mathbf{i} - 91.88\mathbf{j}$

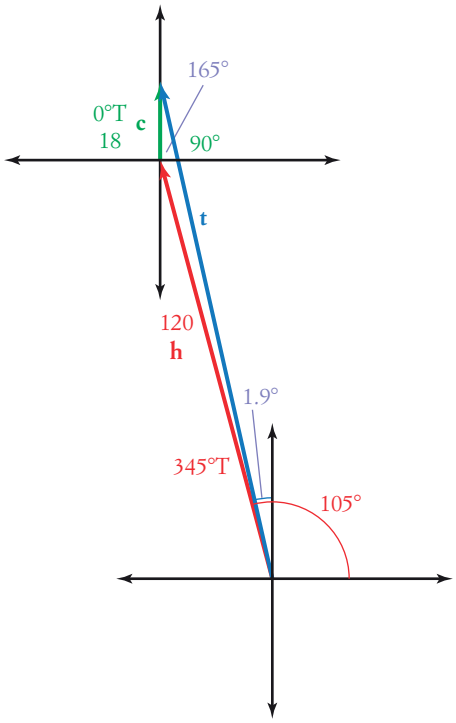
v $|\mathbf{t}| = \sqrt{(-45.92)^2 + (-91.88)^2} \approx 102.71$

vi $\mathbf{T} = 103°$ and $\mathbf{C} = 5.4°$

vii **t** is on a bearing of 210.8°

viii The speed and course made good is about 102.71 knots on a bearing of 210.8°.

d i



ii $\mathbf{h} = 120\cos(105°)\,\mathbf{i} + 120\sin(105°)\,\mathbf{j} \approx -31.06\mathbf{i} + 115.91\mathbf{j}$.

iii $\mathbf{c} = 18\cos(90°)\,\mathbf{i} + 18\sin(90°)\,\mathbf{j} \approx 0\mathbf{i} + 18\mathbf{j}$.

iv $\mathbf{t} = \mathbf{h} + \mathbf{c} = -31.06\mathbf{i} + 133.91\mathbf{j}$

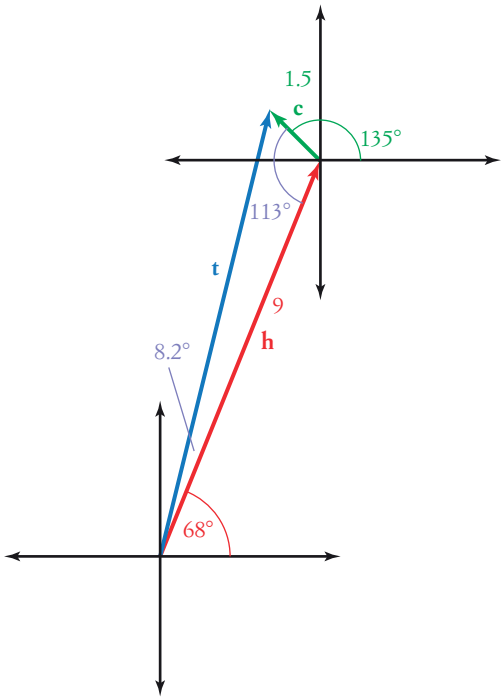
v $|\mathbf{t}| = \sqrt{(-31.06)^2 + 133.91^2} \approx 137.47$

vi $\mathbf{T} = 165°$ and $\mathbf{C} = 1.9°$

vii **t** is on a bearing of 103.1°

viii The speed and course made good is about 137.47 knots on a bearing of 103.1°.

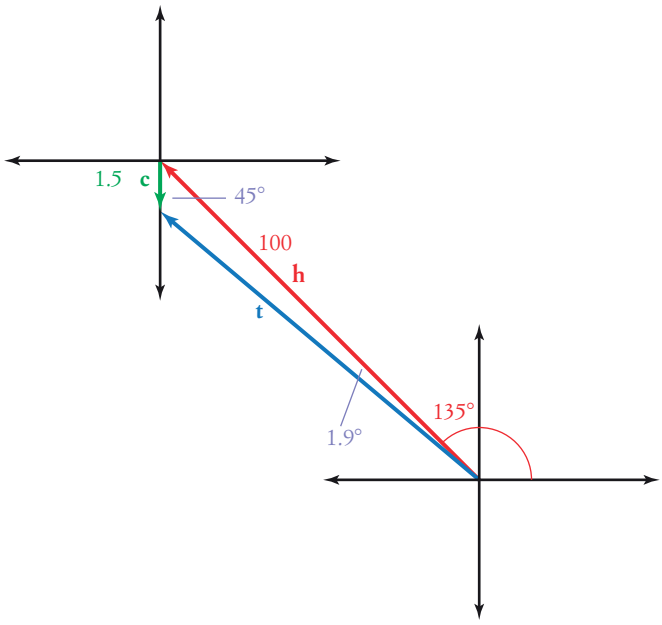
2 a i



ii Speed made good = 9.69 knots

iii Course made good = 13.8°

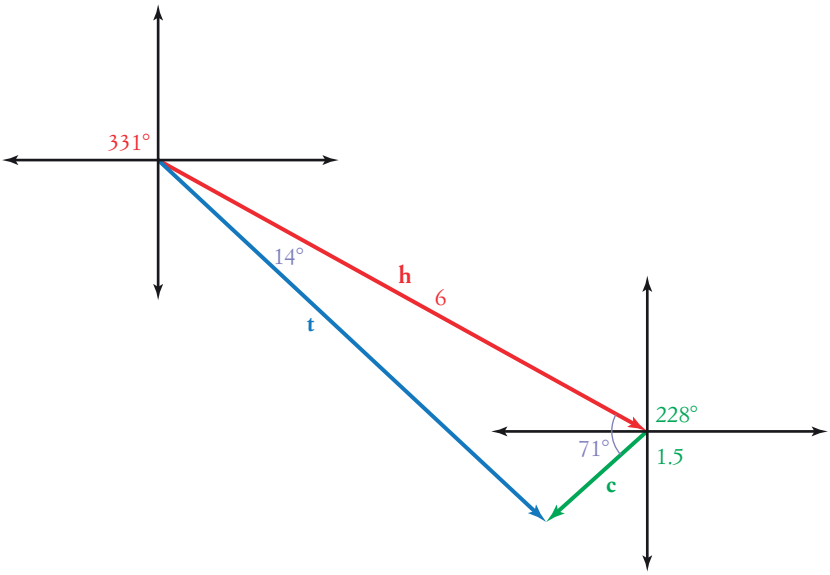
b i



ii Speed made good = 106.52 knots

iii Course made good = 136.9°

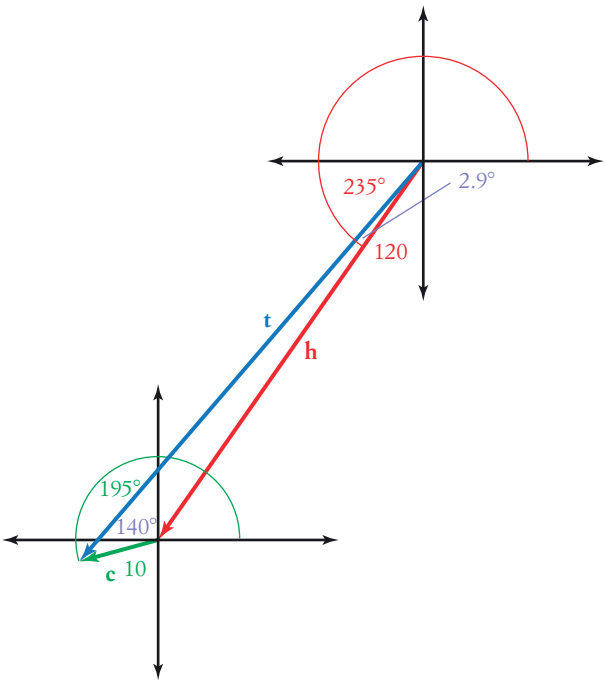
c i



ii Speed made good = 5.85 knots

iii Course made good = 317°

d i



ii Speed made good = 127.82 knots

iii Course made good = 217.9°

3 **a** True course = 354.4° and groundspeed = 130.03 knots

b Speed made good = 11.32 knots and course made good = 306.9°

c Heading = 254.4° and airspeed = 130.03 knots

d Heading = 106.5° and speed through water = 19.26 knots

e True course = 70.2° and speed through water = 10.23 knots

f Speed of wind = 17.4 knots and direction of wind = 296.2°

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